

SEDAR 22: Gulf of Mexico Yellowedge Grouper and Tilefish

Yellowedge Grouper

Data Workshop

LIFE HISTORY WORKING GROUP

- The LH DW recommends directed studies for better estimation of onset of maturity, batch fecundity by age, spawning frequency by age, and spawning duration by age.
- Recommend the fishery-independent longline survey enhance collection of sediment/habitat data to allow post-stratification. Increased resolution of spatial population structure is important given the demographic differences (east and western GOM) noted. There is the potential for over-exploitation of sub-populations within the larger GOM stock.
- Monitor for possibility of increased discards/high-grading as ITQs (catch shares) is undertaken as management approach.
- Since preliminary genetic research and demographic comparisons by Cook (2007) found differences between regions in the GOM the LH DW recommends additional genetic research on population genetics throughout the GOM be conducted.
- Improve information on stock structure/rates of possible exchange between Gulf and Atlantic, including pathways for larval transport.
- Age Johnson historical otoliths collected off Florida during 1982-1983. Use otolith age results to support ages determined using otolith weight to predict age.

COMMERCIAL STATISTICS WORKING GROUP

No recommendations were provided.

RECREATIONAL STATISTICS WORKING GROUP

No recommendations were provided.

INDICES OF ABUNDANCE WORKING GROUP

In both the fishery-independent surveys presented above, precision in abundance indices could be improved by increasing the number of samples at least two- to three-fold.

Research recommendations for fishery dependent data:

1.) Expand observer coverage to provide a subsample adequate to construct indices of abundance (Pelagic Longline Observer Program has 5-8% coverage). Observer data provides finer spatial resolution and a more accurate measure of CPUE. It also provides size frequency and discard information that is currently unavailable in the self-reported dataset. Current observer coverage is inadequate for the construction of indices of abundance.

2.) Self logbook data should be restructured to collect data on a per set basis rather than per trip. This would allow for a more accurate calculation of CPUE. Data subsetting (determining targeting) would be vastly improved with set-based data.

CIE REVIEWER RECOMMENDATIONS - DATA WORKSHOP

Conclusions and recommendations

I would like to commend the great efforts of all the participating scientists, managers and fishermen in the SEDAR 22 DW in the identification, evaluation and compilation of the information on life history, fishery-dependent and fishery-independent abundance indices, and landings in the commercial and recreational fisheries for YG, tilefish (i.e., golden tilefish), and blueline tilefish in the GOM. I was impressed by the breadth of expertise and experience of the panelists, openness of discussion for considering alternative approaches/suggestions, and constructive dialogs in each working group and at the plenary meetings throughout the workshop. All the comments, whether they were from scientists, managers, or fishermen, were fully considered and discussed. In particular, I commend the inclusion in the Data Workshop of fishermen, who provided insights on the quality of the fishery data, in particular for historical fisheries data. I observed on many occasions constructive interactions and dialogs between scientists/managers and representatives of the industry in the Workshop.

In general, I consider the information identified and compiled in the DW represents the best efforts given all the limitations associated with data quality and quantity. I consider the approaches used in developing life history parameters, fisheries landings, and abundance indices sound.

Having said that, I believe that there are large uncertainties associated with data identified and compiled in the DW, and that there is room for further improvement. I have made the following general comments and specific recommendations.

General comments

Although the SoW states that all the working papers and reference/background information for the workshop will be available two weeks before the workshop, only a few working papers (less than 25% of all the working papers promised) were available before the start of the workshop (not mention two weeks before the start of the workshop). Many working papers were still not ready in the middle of the workshop, which made my work difficult. The three separate working groups worked concurrently every day, making it impossible for me, as the only CIE reviewer, to be fully involved in each group's discussions.

I was told at the DW that Stock Synthesis 3 (SS3) will be used for the assessment of YG and tilefish. This choice of stock assessment model has direct impacts on the quality and quantity of the data that need to be evaluated and compiled in the Workshop. However, I observed that most DW panelists did not know exactly the data requirements, key assumptions, and options of the SS3 program. I recommend that future data workshop start with the introduction of the stock assessment model that will be used in the assessment so that data workshop participants understand the information needs of the stock assessment model.

I noticed that the time period that the SEDAR 22 assessment covers had not been defined prior to the DW. I suggest that a stock assessment time period be defined prior to the DW so that working

groups can focus on the defined time period, and not waste time discussing data falling outside the target stock assessment. The DW may also be a good place to discuss and make a decision about the time period the stock assessment should cover.

There is a need to include scientific names for all species covered in the TORs and SoW. The tilefish is the official name of golden tilefish in the American Fisheries Society list of fish species. However, both golden tilefish and blueline tilefish were discussed at the Workshop. This creates some confusion. It is clear from all the discussions at this Workshop that the information for blueline tilefish is not sufficient for a formal stock assessment using an assessment model like SS3.

Specific recommendations

Although I have provided detailed comments and recommendations under each TOR, I re-iterate the following recommendations.

- Possible existence of local stocks for both species needs to be evaluated;
- More comparative studies need to be done to evaluate differences in data collected from different monitoring programs;
- More comparative studies need to be done to evaluate differences in parameters estimated using different methods to improve our understanding of the degree of uncertainty associated with these parameters;
- More comparative studies need to be done to evaluate spatial and temporal variability in key life history parameters, abundance indices and landings;
- More habitat variables need to be included in CPUE and abundance index standardization;
- General additive models need to be considered in standardizing abundance index and CPUE;
- Instead of using a point estimate as a bias correction factor in correcting potential biases in landings data, a range of correction factors can be used so that large uncertainty in landings data can be incorporated into the stock assessment;
- The quality of catch data (landings, catch size/age composition, catch sex ratio etc.) is probably the most questionable of the data available to the stock assessment for both fish species, and the stock assessment model should have an ability to incorporate uncertainty in catch data;
- A critical evaluation of fishery-independent monitoring programs should be done to identify problems associated with the current program design in quantifying population dynamics;
- A systematic mail survey/interview of fishermen who have been involved in the GOM YG and tilefish needs to be done to have a better understanding of the degree of misreporting/underreporting and to identify if there is spatial and temporal variability in underreporting;
- It appears that outliers may exist in the assessment and given the data quality concerns, I suggest that robust estimation methods be used in the assessment (although this may be the choice of the modelers, but I believe that the Data Workshop is a place to make the recommendation because this is the place to deal with data quality issues);
- Uncertainty should be considered in all life history modeling, and confidence intervals should be estimated for the key life history parameters for the GOM YG and tilefish;

- Because of the extremely small YG catch in the SEAMAP bottom trawl survey, caution should be used in applying the derived abundance index, and the change in survey protocol in 1987 calls for a separate analysis of the two time periods and two different catchabilities in population modeling;
- Different measures for SSB should be considered for both tilefish and YG in stock assessment modeling; and
- I recommend conducting a systematic evaluation of current sampling programs for quantifying size composition and age composition of commercial catch. Factors such as adequate spatial and temporal coverage and sampling intensity to have high effective sample sizes should be considered. I recommend developing alternative sampling designs, developing a simulated fishery that mimics temporal and spatial variability in size and age compositions in commercial landings, applying current and alternative sampling programs to the simulated fishery, comparing the performance of the sampling programs with respect to their replications of built-in size and age compositions in the simulated fishery, and identifying a cost-effective port sampling program for quantifying size and age compositions of commercial landings.

Finally, I strongly concur with the recommendations made by the LHG in their draft DW report regarding life history work for the GOM YG and tilefish, and I think all the issues raised in the report are critical to improve the life history data quality. The draft reports of the other two groups (IG and LDG) were not available when I prepared this report so I cannot make any comments regarding the recommendations they will list in the DW reports.

Assessment Workshop

No specific research recommendations were provided.

CIE REVIEW RECOMMENDATIONS – ASSESSMENT PROCESS

The research recommendations in the yellowedge grouper assessment report were all identifying appropriate areas for further investigation but a number of them were rather short on proposed investigative methodologies.

One proposal was to look at genetics. The application of genetics to fisheries management has had mixed success but here is a need to address stock structure and a regional genetics program may be able to address this issue, not only for this species but for others in the same position.

The fishery dependent research recommendations were both good but it is probably worth defining how much observer coverage would be required to provide adequate data from which to construct alternative indices. The additional fishery information obtained from an expanded observer program (on such things as discards) would, however, also be very welcome.

Direct aging of the Johnson otoliths from 1982 and 1983 is a low cost and worthwhile study that will directly feed into future assessments and specifically help to correct the paucity of data in the earlier years of the fishery.

Additional research recommendations have been identified by the reviewer and are presented below in priority order.

Reviewer Recommendations

- In a fishery with multiple data deficiencies, one of the objects of modeling is to identify those data sets that, by their inadequacy or absence, have a disproportionate impact on the outcome of the assessment. This then provides an independent assessment of the prioritization of future research effort aimed at improving the assessment most effectively. More could probably be made of this in defining immediate future research focus.
- Analyze existing data, or collect and analyze new data to confirm that the yellowedge grouper is composed of only a single stock. This could focus on a genetics program aimed at a number of species in the region, as this appears to be a shared problem amongst a number of species.
- Selection bias has occurred in yellowedge grouper age samples, with many more samples in recent years and more from some fishery areas than others (e.g. Florida). Some attempts to obtain a balance of samples from the different areas of (i) the fishery and (ii) the wider stock distribution should be developed and implemented
- While the recreational landings represent a small proportion of the landings it could be worth reviewing the biological data available as recreational fisheries often either target or catch different age or length components of the stock compared to other fisheries. This can be seen in differences between the handline and longline fisheries here. If this is the case then this small part of the fishery may contain useful information about length or age. A basic analysis of length and possibly otolith weight (as a proxy for age) would advise whether this merits further consideration.
- The core input data are in imperial units (lbs) while model processed data (e.g. weight at length or age) are presented in metric units. More importantly the landings/catch data are in lbs and model outputs are in kgs making comparison somewhat difficult. Input and output data should be presented in consistent units.

Review Workshop

The review panel was in agreement with the research recommendations from the Data and Assessment Workshop reports. These identify the main shortcomings in the data and assessment which might be improved by research. However, the recommendations are extensive and some priority may be placed so that research having the greatest impact on the assessment might be given priority.

Based on the observations made during the review, the RP suggested priority might be determined for the following research topics:

1. Research to improve abundance indices and their development from fishery-dependent and fishery-independent data sources would appear to have relatively high priority as they would have a great impact on the assessment. Topics could include, but not be limited to:
 - Improve precision in fishery-independent survey abundance indices by increasing the number of samples, including expansion into deeper water.

- Improve precision in fishery-independent survey abundance indices by expanding observer coverage to at least 5% of the area to provide additional accurate information adequate to construct indices of abundance. Observer data should provide finer spatial resolution, a more accurate measure of CPUE, size frequency and discard information that is currently unavailable in the self-reported dataset. Current observer coverage is inadequate for this purpose.
 - Improve fishery-independent survey abundance indices by using logbooks to collect data on a set-by-set basis rather than per trip. This would allow for a much more accurate calculation of CPUE.
 - Re-examination of the standardization of CPUE indices, both the models and the covariates (habitat, sediment, depth etc.).
2. For yellowedge grouper, ageing could be improved. There are historical otoliths collected off Florida during 1982-1983 which could be used if partitioned between species (e.g. using discriminant analysis). More age data might become available if the relationship between age and otolith weight could be developed. This could have a significant impact the stock assessment.
 3. Research to improve stock definition and structure. For the stock assessment, the biggest impact of this sort of research is on the way data are broken down into areas to try to improve coherence within sub-sets of data. This suggests that priority for this sort of research should depend upon demonstrating that the data can support alternative stock structures and that there would be greater coherence within these subsets of data. There were no apparent cohorts identifiable in the age composition data from the two areas used in this assessment, but insufficient data to support break down into three areas. Improving the basic data through, for example, re-examination of the sampling design for size and age composition from the commercial fishery might have higher priority.
 4. Research on life history is high priority, but should first and foremost be reflected in data collection before assessment model structure. While model structure might be seen as improved in representing real biological processes, such as protogynous hermaphroditism, unless there is sufficient monitoring and other data, the model will effectively be unable to incorporate the process in the assessment. One of the research recommendations which could prove important is to determine a more appropriate way to model spawning stock size for protogynous species.

In addition to research identified in the DW and AW, the RP recommends further work on the stock assessment modelling. The RP found results depended on how different sources of information were weighted, and alternative weighting schemes could be considered in developing future stock assessments. The age and length composition likelihood models appear appropriate, so research may be more focused on the abundance index standardization and ensuring their likelihood model and scale parameters are compatible with the age and length composition likelihood.

The RP also suggested some additional methods which would improve the absolute stock size estimate. These methods would help determine the shape of the selection curve, the value of M, and therefore would improve the MSY estimation. Even though M has been reasonably well estimated, the assessment is still very uncertain, because F and M are low, so further improvements in the estimate of M would be beneficial. Absolute stock estimates might be obtained from 1) underwater video surveys to count fish burrows; 2) deep water tagging, as done for redfish in the Irminger Sea; or 3) depletion fishing experiments within a small area (e.g. 1 x 1 km) combined with NMFS survey type long line fishing to estimate survey catchability, like that done in the REX project for cod and other species in the north-eastern North Sea.

Tilefish

Data Workshop

LIFE HISTORY WORKING GROUP

- In addition to the tilefish reproductive data needs (above), the WG recommends examination of the size frequency of commercial catch by month to examine potential inference concerning the sex ratio of the catch.
- Improve information on stock structure/rates of possible exchange between Gulf and Atlantic, including pathways for larval transport.
- Expand the fishery-independent long-line survey to deeper depths. In addition, increase collection of sediment/habitat data to allow post-stratification of survey results. Increased resolution of spatial population structure is important given the spatially divergent landings and demographic differences (east and western Gulf) and given the potential for localized over-exploitation within the larger Gulf of Mexico stock.
- Last, the WG recommends monitoring the possibility of increased discards/highgrading as ITQs (catch shares) is undertaken as a management approach.

Procedural Recommendation:

At points during the SEDAR 22 process, WG and DW panel members noted some confusion about “tilefish” as a species and as a species complex during discussions. Given the lack of clarity about common names for several species and their associated complexes, The WG recommends that scientific names be added to future SEDAR schedules and announcements.

COMMERCIAL STATISTICS WORKING GROUP

No recommendations were provided.

RECREATIONAL STATISTICS WORKING GROUP

No recommendations were provided.

INDICES OF ABUNDANCE WORKING GROUP

In the fishery-independent survey presented above, precision in abundance indices could be improved by increasing the number of samples at least two- to three-fold.

Research recommendations for fishery dependent data:

1.) Expand observer coverage to provide a subsample adequate to construct indices of abundance (Pelagic Longline Observer Program has 5-8% coverage). Observer data provides finer spatial resolution and a more accurate measure of CPUE. It also provides size frequency and discard information that is currently unavailable in the self-reported dataset. Current observer coverage is inadequate for the construction of indices of abundance.

2.) Self logbook data should be restructured to collect data on a per set basis rather than per trip. This would allow for a more accurate calculation of CPUE. Data subsetting (determining targeting) would be vastly improved with set-based data.

CIE REVIEWER RECOMMENDATIONS - DATA WORKSHOP

Conclusions and recommendations

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In general, I consider the information identified and compiled in the DW represents the best efforts given all the limitations associated with data quality and quantity. I consider the approaches used in developing life history parameters, fisheries landings, and abundance indices sound.

Having said that, I believe that there are large uncertainties associated with data identified and compiled in the DW, and that there is room for further improvement. I have made the following general comments and specific recommendations.

General comments

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I was told at the DW that Stock Synthesis 3 (SS3) will be used for the assessment of YG and tilefish. This choice of stock assessment model has direct impacts on the quality and quantity of the data that need to be evaluated and compiled in the Workshop. However, I observed that most DW panelists did not know exactly the data requirements, key assumptions, and options of the SS3 program. I recommend that future data workshop start with the introduction of the stock assessment model that will be used in the assessment so that data workshop participants understand the information needs of the stock assessment model.

I noticed that the time period that the SEDAR 22 assessment covers had not been defined prior to the DW. I suggest that a stock assessment time period be defined prior to the DW so that working groups can focus on the defined time period, and not waste time discussing data falling outside the target stock assessment. The DW may also be a good place to discuss and make a decision about the time period the stock assessment should cover.

There is a need to include scientific names for all species covered in the TORs and SoW. The tilefish is the official name of golden tilefish in the American Fisheries Society list of fish species. However, both golden tilefish and blueline tilefish were discussed at the Workshop. This creates some confusion. It is clear from all the discussions at this Workshop that the information for blueline tilefish is not sufficient for a formal stock assessment using an assessment model like SS3.

Specific recommendations

Although I have provided detailed comments and recommendations under each TOR, I re-iterate the following recommendations.

- Possible existence of local stocks for both species needs to be evaluated;
- More comparative studies need to be done to evaluate differences in data collected from different monitoring programs;
- More comparative studies need to be done to evaluate differences in parameters estimated using different methods to improve our understanding of the degree of uncertainty associated with these parameters;
- More comparative studies need to be done to evaluate spatial and temporal variability in key life history parameters, abundance indices and landings;
- More habitat variables need to be included in CPUE and abundance index standardization;
- General additive models need to be considered in standardizing abundance index and CPUE;
- Instead of using a point estimate as a bias correction factor in correcting potential biases in landings data, a range of correction factors can be used so that large uncertainty in landings data can be incorporated into the stock assessment;
- The quality of catch data (landings, catch size/age composition, catch sex ratio etc.) is probably the most questionable of the data available to the stock assessment for both fish species, and the stock assessment model should have an ability to incorporate uncertainty in catch data;
- A critical evaluation of fishery-independent monitoring programs should be done to identify problems associated with the current program design in quantifying population dynamics;
- A systematic mail survey/interview of fishermen who have been involved in the GOM YG and tilefish needs to be done to have a better understanding of the degree of

misreporting/underreporting and to identify if there is spatial and temporal variability in underreporting;

- It appears that outliers may exist in the assessment and given the data quality concerns, I suggest that robust estimation methods be used in the assessment (although this may be the choice of the modelers, but I believe that the Data Workshop is a place to make the recommendation because this is the place to deal with data quality issues);
- Uncertainty should be considered in all life history modeling, and confidence intervals should be estimated for the key life history parameters for the GOM YG and tilefish;
- Because of the extremely small YG catch in the SEAMAP bottom trawl survey, caution should be used in applying the derived abundance index, and the change in survey protocol in 1987 calls for a separate analysis of the two time periods and two different catchabilities in population modeling;
- Different measures for SSB should be considered for both tilefish and YG in stock assessment modeling; and
- I recommend conducting a systematic evaluation of current sampling programs for quantifying size composition and age composition of commercial catch. Factors such as adequate spatial and temporal coverage and sampling intensity to have high effective sample sizes should be considered. I recommend developing alternative sampling designs, developing a simulated fishery that mimics temporal and spatial variability in size and age compositions in commercial landings, applying current and alternative sampling programs to the simulated fishery, comparing the performance of the sampling programs with respect to their replications of built-in size and age compositions in the simulated fishery, and identifying a cost-effective port sampling program for quantifying size and age compositions of commercial landings.

Finally, I strongly concur with the recommendations made by the LHG in their draft DW report regarding life history work for the GOM YG and tilefish, and I think all the issues raised in the report are critical to improve the life history data quality. The draft reports of the other two groups (IG and LDG) were not available when I prepared this report so I cannot make any comments regarding the recommendations they will list in the DW reports.

Assessment Workshop

In addition to the recommendations made in the SEDAR 22 Data Workshop Report, the AP makes the following recommendations for research and data collection.

- In a fishery with multiple data deficiencies, one of the objects of modeling is to identify those data sets that, by their inadequacy or absence, have a disproportionate impact on the outcome of the assessment. This then provides an independent assessment of the prioritization of future research effort aimed at improving the assessment most effectively. More could probably be made of this in defining immediate future research focus.
- Analyze existing data, or collect and analyze new data to confirm that the tilefish is composed of only a single stock. This could focus on a genetics program aimed at a number of species in the region, as this appears to be a shared problem amongst a number of species.

- Review the information about distribution of tilefish age in time and geographical area with a view to obtain better quality data going forward (i.e. attempts to obtain a balance of samples from the different areas of (i) the fishery and (ii) the wider stock distribution should be developed and implemented).
- Evaluating whether the amount of remaining quota influences how landings are reported by species should be considered.
- While the recreational landings represent a small proportion of the landings it could be worth reviewing the biological data available as recreational fisheries often either target or catch different age or length components of the stock compared to other fisheries. If this is the case then this small part of the fishery may contain useful information about length or age. A basic analysis of length and possibly otolith weight (as a proxy for age) would advise whether this merits further consideration.

CIE REVIEW RECOMMENDATIONS – ASSESSMENT PROCESS

The research recommendations in the tilefish assessment report were all identifying appropriate areas for further investigation but a number of them were rather short on proposed investigative methodologies.

One proposal to look at stock structure should be part of a larger, probably genetics program to look at regional stock structure in a number of similar species.

The fishery dependent research recommendations were both good but it is probably worth defining how much observer coverage would be required to provide adequate data from which to construct alternative indices. The additional fishery information obtained from an expanded observer program (on such things as discards) would, however, also be welcome.

Additional research recommendations have been identified by the reviewer and are presented below in priority order.

Reviewer Recommendations

- In a fishery with multiple data deficiencies, one of the objects of modeling is to identify those data sets that, by their inadequacy or absence, have a disproportionate impact on the outcome of the assessment. This then provides an independent assessment of the prioritization of future research effort aimed at improving the assessment most effectively. More could probably be made of this in defining immediate future research focus.
- Analyze existing data, or collect and analyze new data to confirm that the tilefish is composed of only a single stock. This could focus on a genetics program aimed at a number of species in the region, as this appears to be a shared problem amongst a number of species.
- Review the information about distribution of tilefish age in time and geographical area with a view to obtain better quality data going forward (i.e. attempts to obtain a balance of samples from the different areas of (i) the fishery and (ii) the wider stock distribution should be developed and implemented).
- Evaluating whether the amount of remaining quota influences how landings are reported by species should be considered.
- While the recreational landings represent a small proportion of the landings it could be worth reviewing the biological data available as recreational fisheries often either target or catch different age

or length components of the stock compared to other fisheries. If this is the case then this small part of the fishery may contain useful information about length or age. A basic analysis of length and possibly otolith weight (as a proxy for age) would advise whether this merits further consideration.

- The core input data are in imperial units (lbs) while processes data (e.g. weight at length or age) are presented in metric units. More importantly the landings/catch data are in lbs and model outputs are in kgs making comparison somewhat difficult. Input and output data should be presented in consistent units.

Review Workshop

The review panel was in agreement with the research recommendations from the Data Workshop and Assessment Workshop reports. These identify the main shortcomings in the data and assessment which might be improved by research. However, the recommendations are extensive and some priority may be placed so that research having the greatest impact on the assessment might be given priority.

The RP noted that the AP suggested using the models to identify the most important data deficiencies to prioritize research and that more work on this could be undertaken. Based on the observations made during the review, the RP suggested priority might be determined for the following research topics:

5. Research to improve abundance indices and their development from fishery-dependent and fishery-independent data sources would appear to have relatively high priority as they would have a great impact on the assessment. Topics could include, but not be limited to:
 - Improve precision in fishery-independent survey abundance indices by increasing the number of samples, including expansion into deeper water.
 - Improve precision in fishery-independent survey abundance indices by expanding observer coverage to at least 5% coverage to provide additional accurate information adequate to construct indices of abundance. Observer data should provide finer spatial resolution, a more accurate measure of CPUE, size frequency and discard information that is currently unavailable in the self-reported dataset. Current observer coverage is inadequate for this purpose.
 - Improve fishery-independent survey abundance indices by using logbooks to collect data on a set-by-set basis rather than per trip. This would allow for a much more accurate calculation of CPUE.
 - Re-examination of the standardisation of CPUE indices, both the models and the covariates (habitat, sediment, depth etc.).
6. Research to improve stock definition and structure. For the stock assessment, the biggest impact of this sort of research is on the way data are broken down into areas to try to improve coherence within sub-sets of data. This suggests that priority for this sort of research should depend upon demonstrating that the data can support alternative stock structures and that there would be greater coherence within these subsets of data. There were no apparent cohorts identifiable in the age composition data from the two areas used in this assessment, but insufficient data to support break down into three areas. Improving the basic data through, for example, re-examination of the

sampling design for size and age composition from the commercial fishery, might have higher priority.

7. Research on life history is high priority, but should first and foremost be reflected in data collection before assessment model structure. While model structure might be seen as improved in representing real biological processes, such as protogynous hermaphroditism, unless there are sufficient monitoring and other data, the model will effectively be unable to incorporate the process in the assessment.

In addition to research identified in the DW and AW, the RP recommends further work on the stock assessment modelling. The RP found results depended on how different sources of information were weighted, and alternative weighting schemes could be considered in developing future stock assessments. The age and length composition likelihood models appear appropriate, so research may be more focused on the abundance index standardisation and ensuring their likelihood model and scale parameters are compatible with the age and length composition likelihood.

The RP also suggested some additional methods which would improve the absolute stock size estimate. These methods would help determine the shape of the selection curve, the value of M , and therefore would improve the MSY estimation. Even though M has been reasonably well estimated, the assessment is still very uncertain, because F and M are low, so further improvements in the estimate of M would be beneficial. Absolute stock estimates might be obtained from 1) underwater video surveys to count fish burrows; 2) deep water tagging, as done for redfish in the Irminger Sea; or 3) depletion fishing experiments within a small area (e.g. 1 x 1 km) combined with NMFS survey type long line fishing to estimate survey catchability, like that done in the REX project for cod and other species in the north-eastern North Sea. This last method may be particularly suitable for tilefish, which is a sedentary species.